

Use Ratio Tables and Unit Rates to Solve Practical Problems



Today I am learning to use ratio tables and unit rates to solve real-life problems.

Step 9

Elementary
(Grades 2–4)

1. Complete the Ratio Tables

Complete each ratio table.

- a) The ratio of apples to oranges is 2 : 3.



Apples	2	4	6	8	10
Oranges	3				

- b) The ratio of boys to girls is 3 : 2.



Boys	3	6	9	12	15
Girls	2				

- c) The ratio of red counters to blue counters is 1 : 4.



Red	1	2	3	4	5
Blue	4				

2. Find the Unit Rate

Find the unit rate (the cost or amount for one item).

- a) 5 pencils cost \$3.50.

Unit rate: \$ _____ per pencil



- b) 4 litres of milk cost \$6.00.

Unit rate: \$ _____ per litre



- c) 12 apples cost \$4.80.

Unit rate: \$ _____ per apple



- d) 6 metres of ribbon cost \$2.70.

Unit rate: \$ _____ per metre



- e) 3 tickets to the zoo cost \$36.

Unit rate: \$ _____ per ticket



3. Use Unit Rates to Solve

Use the unit rate to find the missing amount.

- a) If 5 notebooks cost \$4.50, how much do 9 notebooks cost?

Unit rate: \$ _____ per notebook

9 notebooks cost \$ _____



- b) If 8 metres of fabric cost \$28.00, how much do 15 metres cost?

Unit rate: \$ _____ per metre

15 metres cost \$ _____



- c) If 6 pizzas feed 24 people, how many pizzas are needed for 40 people?

Unit rate: _____ pizzas per person

Pizzas needed: _____



4. Ratio Problems

Solve each problem using a ratio table.

- a) The ratio of dogs to cats in a pet shop is 2 : 5.
If there are 14 cats, how many dogs are there?

Dogs	2			
Cats	5			

Answer: _____ dogs



- b) The ratio of pencils to erasers in a box is 3 : 1.
If there are 12 pencils, how many erasers are there?

Pencils	3			
Erasers	1			

Answer: _____ erasers



5. Real-Life Problems Using Unit Rates

- a) A delivery service charges \$2.75 per kilometre.

How much will a 18 km delivery cost?

Answer: \$ _____



- b) A farmer gets \$1.60 per kilogram for tomatoes.

How much will 37 kg of tomatoes earn?

Answer: \$ _____



- c) A water tank fills at a rate of 15 litres per minute.

How much water will it fill in 8 minutes?

Answer: _____ litres



6. Unit Rate or Ratio?

Decide whether each situation is best solved using a unit rate or a ratio. Write "Unit Rate" or "Ratio".

- a) The cost of 1 litre of juice. _____



- b) The number of red marbles to blue marbles in a jar. _____



- c) The cost for 1 kilometre of taxi travel. _____



- d) The number of girls to boys in a class. _____



7. Challenge Problem

A recipe for lemonade uses 2 cups of lemon juice for every 5 cups of water.

- a) Complete the ratio table.

Lemon juice (cups)	2	4	6	8	10
Water (cups)	5				



- b) If you have 30 cups of water, how many cups of lemon juice do you need?

Answer: _____ cups

- c) If you have 12 cups of lemon juice, how many cups of water do you need?

Answer: _____ cups

Tip!

A unit rate compares to one.

For example:

\$6 for 3 apples

= \$2 per apple

(unit rate)

I can...

- ☐ make ratio tables.
- ☐ find unit rates.
- ☐ solve problems using ratio tables.
- ☐ solve problems using unit rates.
- ☐ explain my thinking.



Key Words

- ratio** – a comparison of two amounts using division
- unit rate** – the rate for one item
- proportion** – two ratios that are equal
- compare** – to look at two or more amounts
- equivalent** – equal in value



Great job!

You are using ratio tables and unit rates to solve practical problems!